Jackson Avery *Curriculum Vitae*

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EDUCATION

M.A., Education Curriculum & Instruction, *University of Colorado Boulder* *May 2024*
Capstone: "*From Bias to Belonging: Dismantling Stereotypes & Supporting Science Identity in High School Science"*
Advisors: Will Lindsay, Ian Her Many Horses, Valerie Otero
Licensure: Science-Secondary Education
GPA: 4.0/4.0

B.A, Astronomy (Astrophysics/Physics), *University of Colorado Boulder* *May 2023*
Degree Honors: *Phi Beta Kappa, With Distinction*
Minors: Space, Education
Certificate: STEM Education
GPA: 3.973/4.0; Class Rank: 2/76

HONORS AND AWARDS

Northglenn High School
Young Educator of the Year *May 2025*

University of Colorado Boulder School of Education
Robert Noyce Scholar **($18,500)** *May 2023*

University of Colorado Boulder College of Arts & Sciences
Jacob Van Ek Scholars Award **($230)** *April 2023*

Department of Astrophysical and Planetary Sciences
Astrophysical and Planetary Sciences Perspectives Scholarship **($1,500)** *May 2022*
Richard A. McCray Scholarship **($3,000)** *April 2021*

PUBLICATIONS

[In Preparation] **Avery, J.**, Berta-Thompson, Z. K., Stephens, A., & Smith, R. (2025). *Design Considerations for Ground-Based Slitless Spectrophotometry.*
Target Journal: American Astronomical Society (AAS)

[In Review] Berta-Thompson, Z. K., Wachiraphan, P., Stephens, A., Caradonna, M., Murray, C., Arriero, V., **Avery, J.**, Duvvuri, G. M., & Pineda, S. (2025). *exoatlas: friendly Python code for exoplanet populations.* arXiv preprint arXiv:2507.02210.
Submitted to Journal of Open Source Software (JOSS)

RESEARCH EXPERIENCE

Research Assistant, ***University of Colorado Boulder & Sommers-Bausch Observatory***
Project: Ground-Based Slitless Spectrophotometry *June 2025 - Present*
Advisor: Zach Berta-Thompson

- Validating theoretical noise models with observational data to assess the potential for measuring statistically significant transit depths using ground-based slitless spectrophotometry.
- Developing Python pipelines and analysis methods to enhance slitless spectrophotometry precision for accessible research and educational applications.
- Characterizing instrumental efficiency and quantifying observational challenges, including additional noise contributions from sky brightness, atmospheric scintillation, and crowded observation fields.

Graduate Student Researcher, ***University of Colorado Boulder*** *January 2024 - May 2024*
Project: Advancing Inclusive Science Education Through Identity-Oriented Curriculum
Advisors: Will Lindsay, Ian Her Many Horses, Valerie Otero

- Designed & implemented an inclusive physics curriculum, elevating contributions from marginalized scientists and recontextualizing scientific history to investigate effects on students' identity development as scientists.
- Applied mixed-methods analysis, including coded Draw-a-Scientist Tests, pre/post identity surveys, and qualitative reflections, to quantify shifts in students' perceptions as scientists and belonging in STEM.
- Proposed evidence-based recommendations for equity-focused science pedagogy emphasizing identity, representation, and inclusivity in physics education.

Undergraduate Research Project, ***University of Colorado Boulder*** *(PHYS/EDUC 4460)*
Project: Comparative Analysis of Introductory Astronomy Labs *February 2023 - May 2023*
Advisors: Noah Finkelstein, Seth Hornstein

- Analyzed 22 introductory astronomy lab activities to compare knowledge types, cognitive demands, and question distributions for science vs. non-science majors, finding majors' labs emphasized higher-order thinking while non-majors' focused on lower-order factual-objective skills.
- Implemented and adapted the Colorado Learning Attitudes about Science Survey from physics to astronomy, revealing more expert-like attitudes among science majors and notable gender disparities within both major and non-major courses.
- Presented findings at the course poster session and wrote a term paper highlighting opportunities to enhance higher-order thinking and address metacognitive gaps in undergraduate astronomy lab curricula.

Undergraduate Research Project, *University of Colorado Boulder* *(ASTR 3510)*
Project: Transit of the Exoplanet HAT-P-53 b *November 2022 - December 2022*
Advisor: Zach Berta-Thompson

- Collected and analyzed ground-based transit observations of the hot Jupiter HAT-P-53 b to measure its transit depth using differential photometry with a comparison star to correct for variable seeing conditions.
- Developed a Python pipeline for calibration, exposure alignment, and differential photometry to extract time-series transit light curves.
- Presented results at a course poster session, including transit modeling, uncertainty analysis, and recovery of the accepted transit depth and planet radius.

TEACHING EXPERIENCE

Secondary Science Teacher, *Northglenn High School* *Fall 2024 - Present*
Courses: General Physics, Geology

- Develop and implement daily lessons, labs, and assessments for ~130 students, aligned with Next Generation Science Standards and Colorado Academic Standards at a high-needs STEM school.
- Partner with *CU-STARs (University of Colorado Boulder)* to coordinate outreach events, including pre-visit planning, day-of classroom support, and post-visit assessments to bridge secondary and post-secondary student opportunities.

Instructor, *University of Colorado Boulder & PEER Physics* *Summer 2024*
Course: Energy and Interactions *(PHYS/EDUC 1580)*

- Co-instructed an introductory physics course for 7 non-physics undergraduates, facilitating active learning and guiding hands-on experiments.
- Designed and implemented inquiry-based labs on planetary magnetic fields, inverse-square law, solar energy, and impulse–momentum theorem as part of the PEER Physics curriculum.

Learning/Teaching Assistant, *University of Colorado Boulder* *Spring 2021 - Fall 2023*
Courses: Introductory Astronomy 1 *(ASTR 1010);* Accelerated Introductory Astronomy 1 *(ASTR 1030)*; Formation & Dynamics of Planetary Systems *(ASTR 3710)*; Planets, Moons, and Rings *(ASTR 3750)*; Becoming a Learning Assistant *(EDUC 4610)*

- Assisted in lectures (~100-200 undergraduates) and labs (~20 undergraduates) for introductory and advanced astronomy/planetary science courses.
- Supported student engagement, guided labs, and applied equity-focused STEM education strategies to foster deeper learning across undergraduate astronomy courses.

Outreach Presenter & Navigator, *Fiske Planetarium* *September 2022 - June 2023*

- Created inclusive, engaging planetarium presentations on the solar system and planetary formation, tailored to diverse K–12 and public audiences.
- Led hands-on astronomy outreach events, including presenting & navigating planetarium shows, kinesthetic astronomy, and meteorite demonstrations.

RELEVANT SKILLS

- Python & Jupyter
- LaTeX & Overleaf
- Telescope Operations
- Data Visualization
- Signal-to-Noise Estimation
- Science Communication
- Scientific Writing
- STEM Outreach
- Collaborative Research & Teaching
- Inclusive Pedagogy